



AI-BASED SOLUTION FOR LEARNING DISABILITIES



Multimodal Dyslexia Dataset Collection Platform for Indian Students

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Abstract

Dyslexia affects a large student population in India, yet existing learning-support technologies rely on Western datasets that fail to capture the linguistic and cultural patterns of Indian learners. To address this, we developed a **secure, web-based platform** designed to collect multimodal data like **reading, writing, speech, typing, and keystroke dynamics** from Indian school children. This semester, the platform was migrated to **AWS**, deployed publicly at **dyslexia.flamenlp.in** with HTTPS, expanded with NCERT-aligned tasks, synchronized audio-video recording, and real-time performance statistics. An Android APK extends access to mobile devices. The platform is now production-ready for controlled school-based deployment.

Methodology

- 01 Cloud Deployment on AWS** Migrated to AWS: Flask on EC2, MySQL on RDS, and media assets in a private S3 bucket. Acquired domain dyslexia.flamenlp.in with full SSL/HTTPS encryption platform-wide.
- 02 Expanded Multimodal Task Engine** Task library scaled to 6–12 NCERT-aligned tasks per class across 7 types: Reading Aloud, Writing, Typing, Reading Comprehension, Aptitude, Listening Comprehension, and Visual Aptitude – all verified against Bloom's Taxonomy by domain specialists.
- 03 Synchronized Audio-Video Recording** Browser-native A/V capture with a three-choice consent flow (video+audio / audio-only / decline), live camera preview widget, and independent per-stream toggle controls. Combined files uploaded to S3 on submission.
- 04 Performance Statistics & Feedback** Post-task statistics display reading accuracy, WPM, typing speed, and normalized scores. Badges awarded for milestones. A five-star post-task feedback widget captures student experience for iterative improvement.
- 05 Accessibility, Legal Compliance & Android APK** Accessibility suite (Reading Ruler, OpenDyslexic font, Monochrome Mode) retained platform-wide. Terms of Service, Privacy Policy, and parental consent form finalized with legal review. Android APK packaged for mobile access.

Results

This semester delivered a fully functional, production-grade platform live at **dyslexia.flamenlp.in**, with **stable AWS infrastructure** handling concurrent submissions across all user roles with no data loss, no integrity violations, and no downtime.

Post-task performance statistics computed correctly for all task types. **Camera and microphone** consent validated across Chrome, Firefox, and **Android** devices. The task library now hosts 6–12 **NCERT-aligned tasks** per class with all task types integrated into the **save-submit-retake** engine. The Android APK ran successfully across three tested device models. These results confirm **production readiness** for controlled school-based deployment.

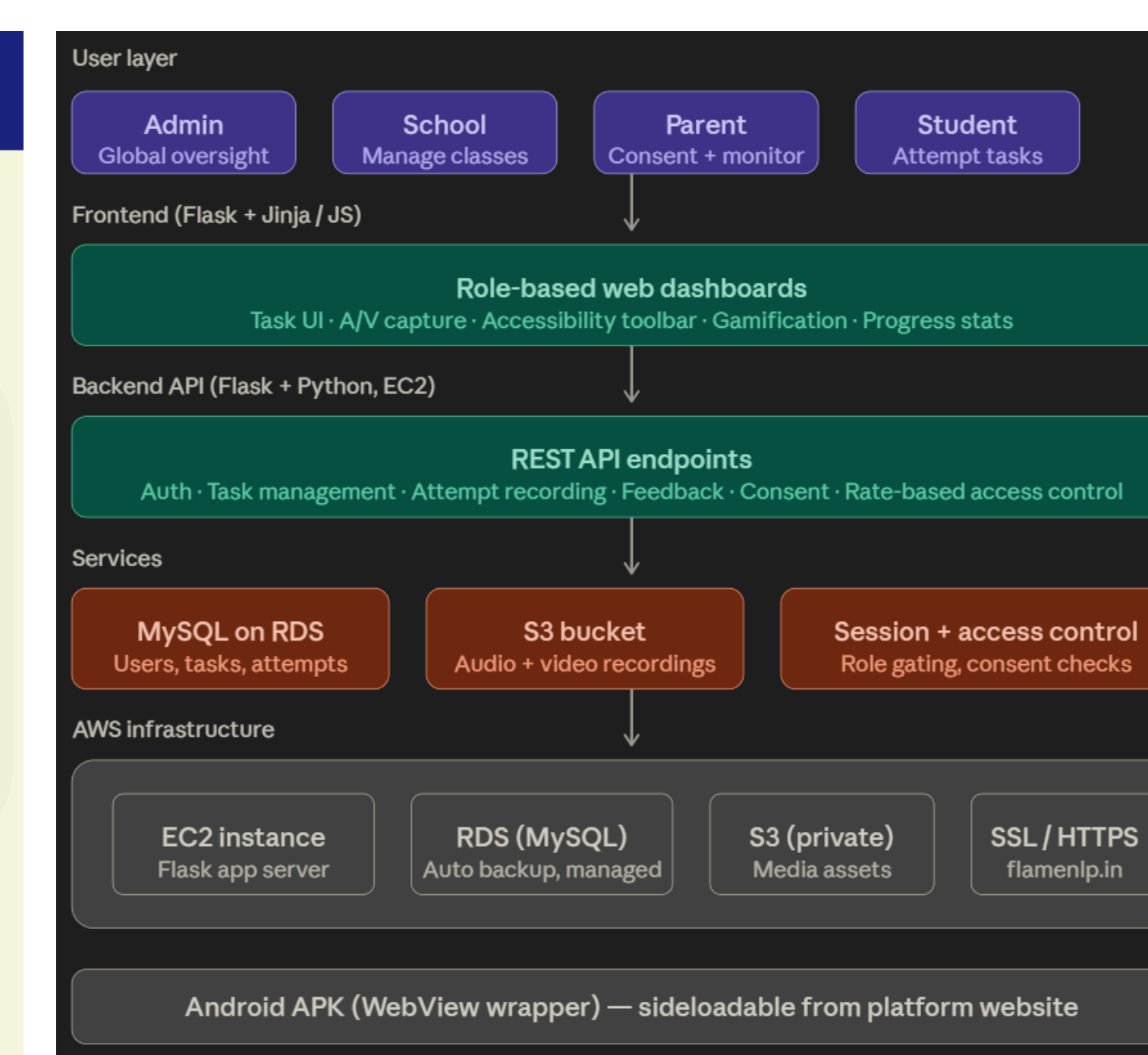
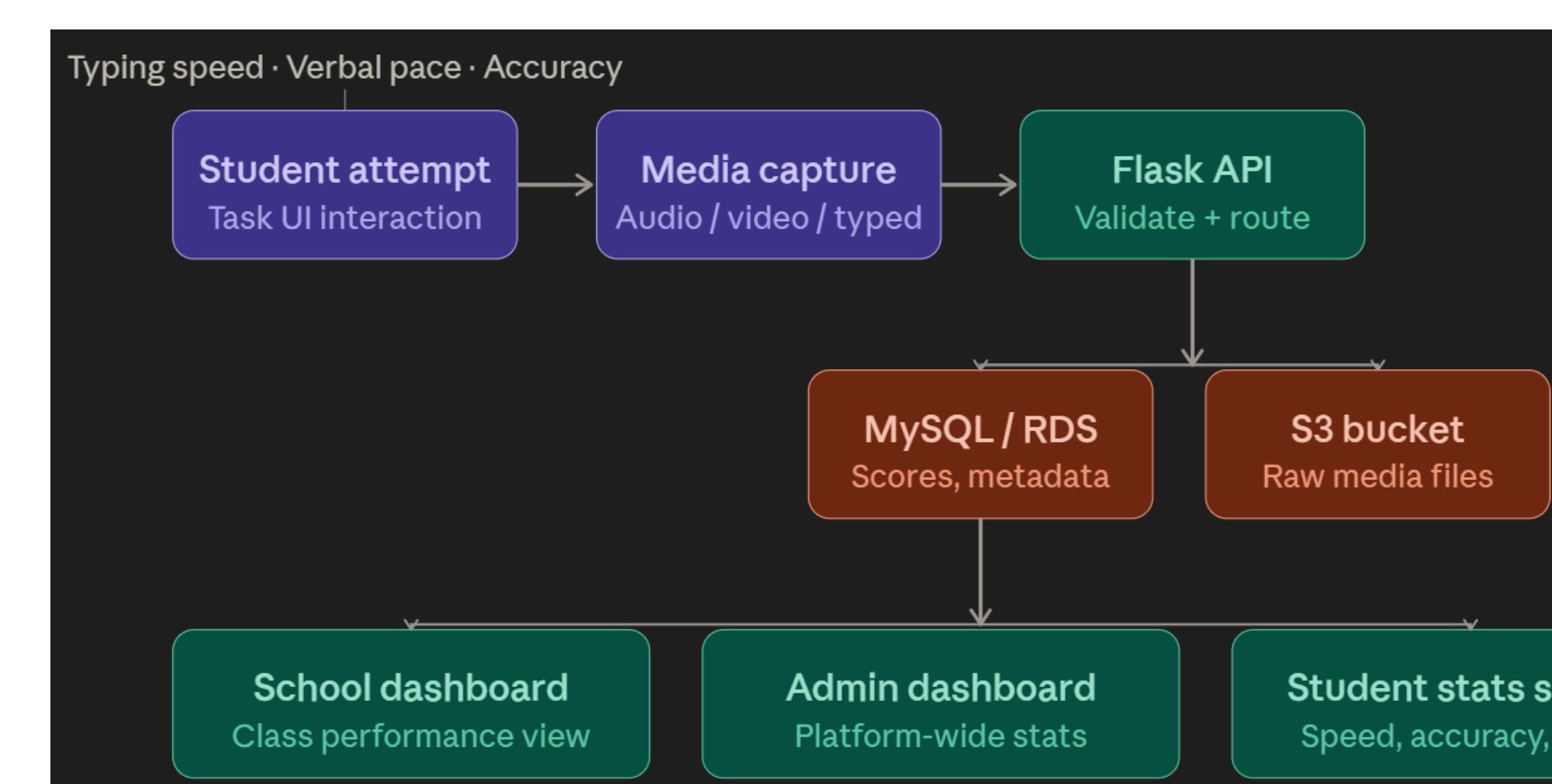
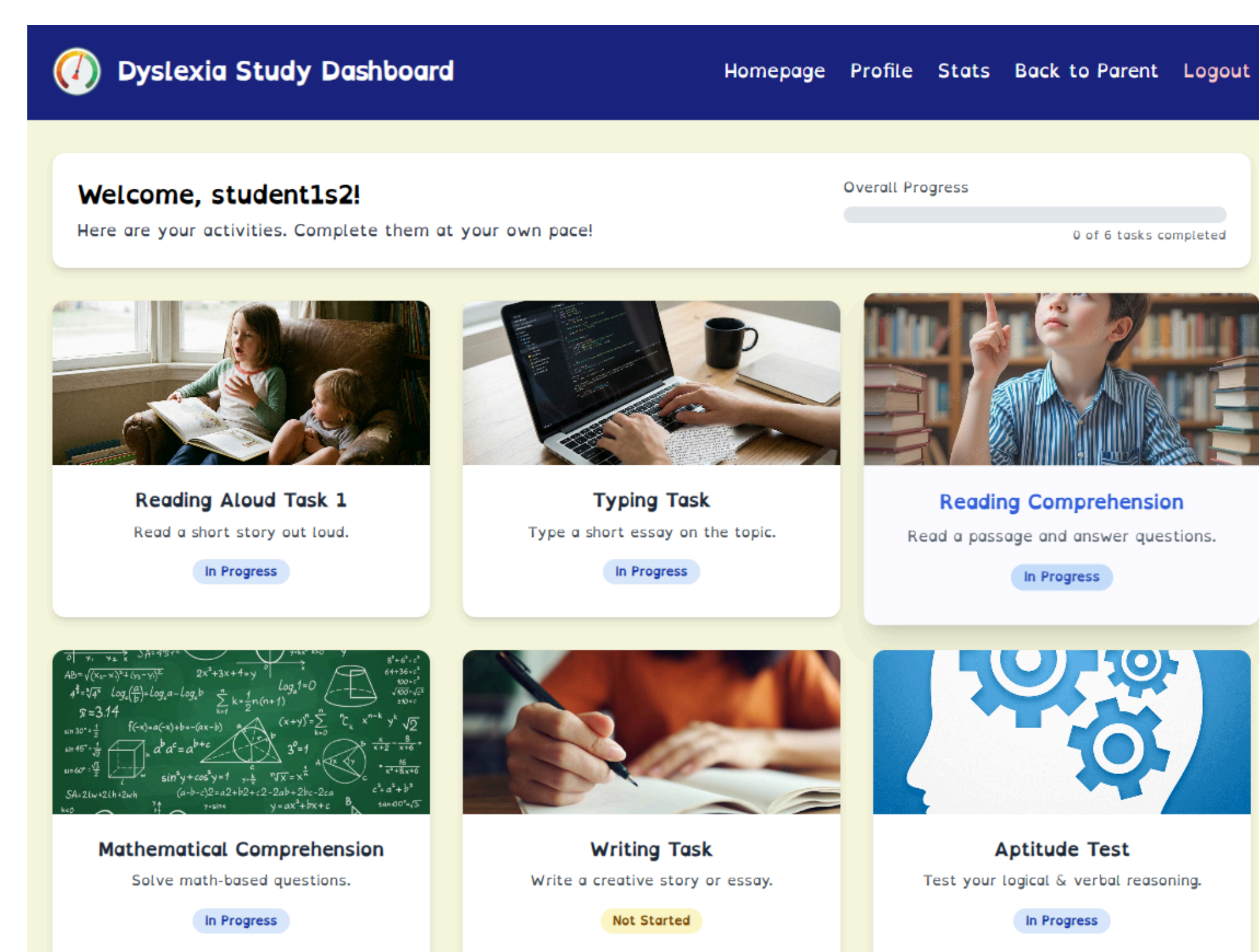
Discussion

This platform shifts dyslexia assessment from passive form-filling to **active, behavior-rich multimodal engagement** at production scale. Synchronized audio-video recording substantially enriches the behavioral signal-capturing facial expression, and posture beyond what audio alone provides. The NCERT-aligned task library ensures collected data reflects the **core limitation of Western datasets**. The "Give and Take" model is now fully realized: students receive real-time performance **feedback motivating re-engagement**. This platform forms the foundation for **Indian-context AI models for automated screening, early detection, and personalized intervention**.

Introduction

Specific Learning Disabilities (SLDs) such as Dyslexia affect a significant portion of the Indian student population. However, current automated screening tools are predominantly trained on **monolingual Western datasets**, rendering them ineffective for the multilingual and phonetically distinct Indian context.

This project addresses the critical **"data scarcity gap"** by engineering a robust infrastructure for data acquisition. Semester 1 delivered the complete technical foundation. This semester transformed it into a **production-grade, publicly hosted, legally compliant platform at dyslexia.flamenlp.in**. The primary objective is to build a large-scale annotated repository to facilitate the **development of culturally relevant AI assessment tools**.



Conclusion & future work

We transformed a **locally validated prototype** into a **fully functional, scalable, production-grade multimodal data-collection platform** for Indian students with dyslexia. This semester delivered AWS cloud deployment, HTTPS via a custom domain, 7-task-type NCERT-aligned library, synchronized audio-video recording, performance statistics, legal compliance, and an Android APK.

Future work will deploy to partner schools for real-world data collection, expand tasks to Hindi and regional languages, build an SPL educator annotation dashboard, develop ML screening models using audio, keystroke, and comprehension features, and release a publicly available, ethically licensed Indian multimodal dyslexia dataset to the research community.